# 9.4 The Tangent Ratio

Name _____ Period ____ Date _____

Geometry 9.4 (30-minute plan) · Student sheet

**I can…**

- find the tangent of an acute angle in a right triangle.
- use tangent to find a missing leg of a right triangle.
- use an angle of elevation to solve real-life problems.

**Vocabulary** · fill in the missing word

**Trigonometric ratio** — a ratio of the lengths of two sides of a right _____

**Tangent** — tan A = (leg _____ ∠A) ÷ (leg adjacent to ∠A)

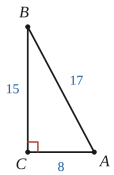
**Angle of elevation** — the angle from the _____ up to your line of sight

## Part 1 · Notes and examples (fill in each note, then work the example)

**Tangent:** tan A = _____ ÷ _____

### Example 1

Find tan A and tan B. Write each as a fraction and as a decimal rounded to four places.

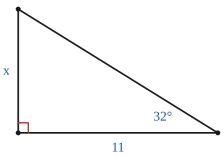



**Same angle, same ratio:** the tangent of an angle is the same in every _____ right triangle, no matter how big.

**Finding a leg:** write tan = opposite ÷ adjacent, put in what you know, and _____ for x. Keep the calculator in _____ mode.

### Example 2

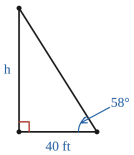
Find the value of x. Round to the nearest tenth.




**Angle of elevation:** the angle from the _____ up to your line of sight.

### Example 3

You stand 40 feet from the base of a tree. The angle of elevation from the ground to the top of the tree is 58°. How tall is the tree? Round to the nearest tenth.



## Part 2 · Partner practice

### Partner practice

Find the value of x. Round to the nearest tenth.

